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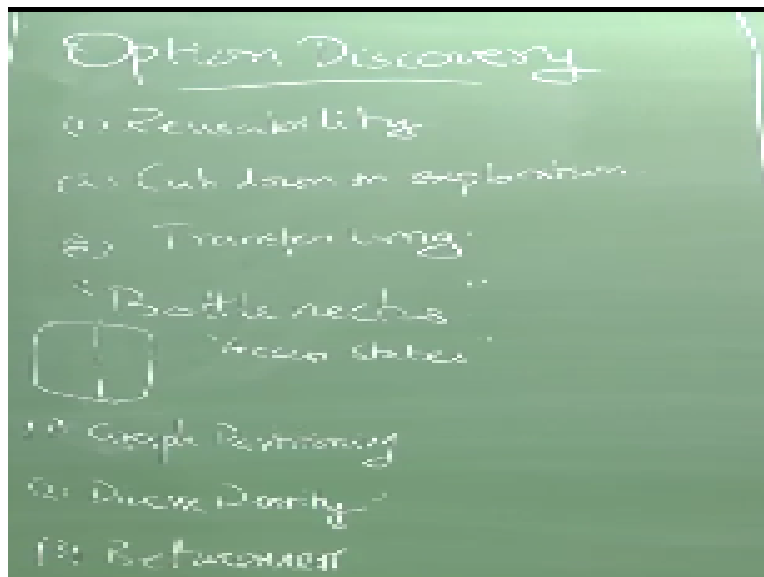
REINFORCEMENT LEARNING

Option Discovery

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I want to very quickly talk about option discovery right so I mean forget about how we define max Q hierarchies and things like that I mean the more complex the hierarchical formulation the harder it becomes to automatically discover them right but with options it is kind of easy right so you can actually talk about option discovery rather in a straightforward fashion.

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So what would be a good option for you to discover when do you call an option good okay so one thing is reusability okay second thing would be I would say that it cuts down on cuts down on exploration right allows you to learn faster right and it aids in transfer learning but turns out that all of these are actually hard ways to evaluate how good an option is right if I want to say it cuts down on exploration and other things how will you how will you evaluate it or how will you find out an option that cuts down on exploration it is a little tricky right.

you have to run multiple times like with one set of options then you have to pick another set of options run multiple times to see if it cuts down exploration and things like that so what people typically tend to do is instead of looking at the actual thing you want from options they try to use some kind of a surrogate function right like surrogate measures so what could be a surrogate measure that they use so the first thing that they use this is what is called bottlenecks.

So what are bottlenecks something that is limiting that is a bottleneck right so you have a bottle and the necks are bottlenecks are usually narrow right so something that kind of has like this funnel kind of an effect there are lot of things but then they have to all pass through a very narrow constriction to reach from one point to another right so for example have you seen any bottlenecks in any of the examples you have seen so far yeah doorways right even that simple to have problem that we looked at earlier right.

The doorways are actually bottlenecks because to go from one one half of the MDP to the other half of the MDP we had only two states which acted as this bottlenecks are sometimes called the sometimes called access states because they allow access to some other part of the MDP right so people said we are going to find bottlenecks so why bottlenecks why are bottlenecks a useful heuristic to have well they obviously will cut down on exploration because if you know how to get to through this door very effectively then I can explore from here I can go to the states here very easily.

Otherwise if I am trying to do random exploration I have to hit those just exactly those two states to get from one room to another so it makes it hard so it cuts down on exploration great what about reusability yeah even if I keep moving the goal around I will probably have to go through

this so it is probably something that I am going to reuse a lot even while learning right and also for solving multiple problems what about transfer learning again the same thing if I move the goal around also knowing where the bottleneck states are will be useful.

So essentially people said oh okay so bottlenecks are good things to find right so how do you find bottlenecks so depending on the level of information that we have about the problem right if I give you the p right if I give you the transition structure of the MDP how will you find out bottlenecks sorry great that is essentially what you do so essentially you try to segment the MDP we can model it as a graph right so two states are connected if there is an action that will take me to take you from one state to another right.

Now you model this as a graph and then try to find graph cuts right find try to find the components of the graph which are very weakly connected and then those states where the weak connection happens would be your bottleneck states right so there are several approaches that actually use this basic idea and yeah so some kind of I am going to say there is some kind of some kind of graph partitioning ideas right so there is work by Shie Mannor way back 2003 or 2002, 2001 or something.

On where they do this kind of graph clustering to find the connected components not connected components graph clustering to find the different components and then do it so we have done some work on using graph partitioning to find options right and it is all of what Ramnandan is doing now is essentially this right so a lot of lot of approaches for doing this okay what if you did not have this graph what if we did not have the MDP how will you do it one simple thing is to say do a lot of just walk around randomly in the MDP collect data from the MDP and then do a graph partitioning is there anything else that you can do.

Turns out that you can do even simpler things right so one of the simplest thing is to look at what people call diverse density so what did they do in the diverse density approach is they are going to assume that you have many trajectories that you have generated solving a problem in the MDP you have many many trajectories that you have generated solving a problem in the MDP right and many many trajectories that you have generated where you have aborted you have not the reached the goal right.

you have successful trajectories and you have unsuccessful trajectories right so you look at states that appear a lot on the successful trajectories but do not appear a lot on the unsuccessful trajectories again it is a heuristic but the heuristic essentially tells you that ok this is the state that I needed to get through to reach to the goal right if the times when I did not make it through this state I never reach the goal right but it is also possible.

For example in the rooms case it is possible that I came from one room to the other but still I aborted because I was taking too long in the other room that is also possible right but that is why I am saying it is not 0 right in the unsuccessful trajectories also the state might occur but it occurs disproportionately more in the successful trajectories you have to keep track of the trajectories I need a lot of successful trajectories and a lot of unsuccessful trajectories and then on those successful trajectories I count I keep track of all the trajectories all the trajectories I have seen so far a few hundred few thousand trajectories how many ever you want right.

And from there you can do this is called the diverse density approach so you it has more often occurred in the successful trajectories and less often in the unsuccessful trajectories right in fact that was the first option discovery paper that was written and unfortunately people have not given too much thought into more principled approaches to discovering options right and so that turned remains one of the most implementable approaches because it requires I mean however few data you give it however little data you give it it can throw back some options at you right.

This kind of graph partitioning approaches you need a good model of the MDP before you get reasonable options out of it and people have not really looked at you know what happens if you are having ill specified model and so on so forth we are doing some work on it now so hopefully something interesting would come out of it soon right and the third approach is essentially it is like graph partitioning right.

I would say it is still a graph partitioning approach even though the author would disagree with that it is based on betweenness right so betweenness is a concept from the network analysis community where a node has a high betweenness if a lot of shortest paths pass through that node

right so I take a pair of vertices on the graph right I find the shortest path between the pair of vertices I find all such shortest paths right there could be many shortest paths.

If suppose you take a grid right I want to go from one point to another I can go any combinations of lefts and rights right as long as the it covers the Manhattan distance between the two points that will be a shortest path right betweenness is a peculiar centrality measure that allows you to find bottlenecks in fact there is a very strong correlation between betweenness and bottlenecks in fact you can define bottlenecks through betweenness right.

So really high betweenness means that that not necessarily that all the other centrality measures will yield a direct analogue for example degree centrality does not necessarily mean it is a bottleneck right but betweenness centrality means it is a bottleneck high betweenness centrality means it is a bottleneck right so essentially what you do is you run your betweenness centrality algorithms on the graph and then find the states with high betweenness centrality and then call them as your bottleneck states once you find bottleneck states.

you have ways of finding options defining options you need π and β so the bottleneck states give you the β right bottleneck states give you the β they will tell you where you want to stop right but which are the appropriate states to start from what is the policy to follow to get there all of those things must be determined right so in the diverse density case they do it very naturally by using experience replay.

So I have so many trajectories from which I have learnt figured out which are the bottleneck states right I just replay those trajectories and learn the q functions for going to the option policy for going to those bottleneck states right and in fact were so we did some we did a fun project in the social networks course when I was teaching where we looked at random options which we call small world options essentially we inserted options at random into the world but the expected length of transition of the options followed a certain probability distribution okay.

Which is what was needed to convert the MDP into a small world MDP small world is essentially you have you reach from any node to any other node in a small number of hops right

especially $\log n$ hops right so you want to reach from any point to any other point so n is the size of the graph I want to be able to go from any point to any other point on the graph in $\log n$ hops actually $\log n$ squared hops $\log n$ into $\log n$ hops right so that is a short that is a small number in comparison with n right so that is called a small world graph and there are results that say that if you manage to insert edges into the graph that have a certain length distribution you can convert them into a small world graph.

So essentially we just threw in small world options and then showed that that cuts down the exploration time significantly right so it cut improves the learning time significantly so for these you do not really need any prior experience or anything you do not need to have even solved the problem right all you need is some random trajectories and any number of trajectories random trajectories will give us we can define useful options out of it.

So in fact we have a data equivalence test so I give you a certain number of experiments that you can do okay with that data that you gather you can compare against the betweenness kind of a method right and also a graph partitioning kind of an approach and see which one makes better use of the data it turns out the small world options makes best use of the data therefore giving a better way.

So that is a completely off the completely off the wall thing for trying to discover this of course I am not sure how well it will do in transfer learning but it should do in well in transfer learning because it only worries about the dynamics of the problem it has nothing to do with the reward functions right so the small world option so it should do well in transfer learning it certainly cuts down on exploration and certainly has a high reusability because it just connects different points in the world right so these are these are some of the approaches that people have out there so I will stop here so this is all about hierarchies.

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